

Amplitude and Probability

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Reading Assignment

Scott Aaronson - Different Rules of Probability

- ▶ <https://www.quantamagazine.org/why-is-quantum-computing-so-hard-to-explain-20210608/>

The screenshot shows the top of a web article. At the top left is a logo of a network of nodes. Next to it is the title "What Makes Quantum Computing So Hard to Explain?". To the right of the title are icons for comments (22), a bookmark, and a share button labeled "SHARE". Below the title is a horizontal line. On the left side, there is a circular profile picture of Scott Aaronson, followed by his name "Scott Aaronson" and the title "Contributing Columnist". Below that is the date "June 8, 2021". Underneath the date is a link "VIEW PDF/PRINT MODE" with a small icon. Below this is a list of tags: "computational complexity", "computer science", "physics", "Quantized Columns", "quantum computing", "traveling salesperson problem", and "All topics" with a right arrow. To the right of the tags is a "Watch" button above a video player showing a blue and yellow abstract pattern. The main text of the article begins with a large "Q" and reads: "uantum computers, you might have heard, are magical uber-machines that will soon cure cancer and global warming by trying all possible answers in different parallel universes. For 15 years, on my blog and elsewhere, I've railed against this cartoonish vision, trying to explain what I see as the subtler but ironically even more fascinating truth. I approach this as a public service and almost my moral duty as a quantum computing researcher. Alas, the work feels Sisyphean: The cringeworthy hype about quantum computers has only increased over the years, as corporations and governments have invested billions, and as the technology has progressed to programmable 50-qubit devices that (on certain contrived benchmarks) really can give the world's biggest supercomputers a run for their money. And just as in cryptocurrency, machine learning and other trendy fields, with money have come hucksters." Below this is a paragraph starting with "In reflective moments, though, I get it. The reality is that even if you removed all the bad incentives and the greed, quantum computing would still be hard to explain briefly and honestly without math. As the quantum computing pioneer Richard Feynman once said about the quantum electrodynamics work that won him the Nobel Prize, if it were possible to describe it in a few sentences, it wouldn't have been". On the right side of the article, there is a section "Share this article" with icons for Facebook, Twitter, LinkedIn, and Email. Below that is a "Newsletter" section with the text "Get Quanta Magazine delivered to your inbox" and a "Subscribe now" button. At the bottom of the newsletter section is a link "Recent newsletters" with a right arrow.

What Makes Quantum Computing So Hard to Explain?

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Scott Aaronson
Contributing Columnist
June 8, 2021

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